Mathematics Bootcamp — One Statement, Three Proofs

Direct, contrapositive, and contradiction on a single claim

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Companion to Day 1: Formal Language and Proof Structure

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One claim, proved three times. The arithmetic is deliberately trivial, so that nothing on the page competes with the thing being taught: what changes between the three methods is not the mathematics but the *bookkeeping* — what you are allowed to assume, and what you are obliged to produce.

The claim

Claim. Let $n \in \mathbb{Z}$. If n is even, then $n + 3$ is odd.

Write

$$P: \text{ ``}n\text{ is even''}, \qquad Q: \text{ ``}n+3\text{ is odd''},$$

so the claim is $P \to Q$, to be proved for every $n \in \mathbb{Z}$.

Definition. An integer n is *even* if $n = 2k$ for some $k \in \mathbb{Z}$, and *odd* if $n = 2k+1$ for some $k \in \mathbb{Z}$.

Read the definition as a contract in both directions. To *use* "n is even" is to obtain a k; to *prove* "n is even" is to exhibit one. Every step below is one of those two moves, an application of a background fact, or ordinary algebra.

The only arithmetic assumed: $\mathbb{Z}$ is closed under addition, subtraction and multiplication.

Two background facts

The three proofs do not all rest on the same ground, and it is worth knowing which fact each one spends. Both facts below are about parity alone; neither mentions the claim, and both are taken as given here.

Fact 1 (every integer has a parity). Every $n \in \mathbb{Z}$ is even or odd.

Fact 2 (no integer has two parities). No $n \in \mathbb{Z}$ is both even and odd.

Neither is proved here. They are the case $d = 2$ of the statement that dividing by d leaves exactly one remainder in $\{0, \ldots, d-1\}$, which is where they belong; assume them today and take them up when the tools are available. Fact 1 in particular is the sort of "obviously true" statement whose proof needs induction, which we have not reached yet — a good illustration that obvious and easy to prove are different properties.

What matters here is only how they get *used*:

(1) Fact 1 turns "n is not odd" into "n is even" — a negative hypothesis, which offers nothing to compute with, into a positive one that hands you a k;

(2) Fact 2 turns "n is odd" into "n is not even" — letting a negative conclusion be reached by proving a positive statement.

Between them, "not odd" and "even" are interchangeable, and so are "not even" and "odd".

Remark. Keep an eye on where these get used below. Fact 1 appears twice — once in each of Proofs 2 and 3, in both cases on the very first line — and never in Proof 1. That is not an accident, and it is the point the three proofs together are making.

Why there are three methods at all

Each method is licensed by a logical equivalence, and each equivalence is a column of a truth table that the Day 1 slides already carry:

P	Q	$\neg P$	$\neg Q$	$P \to Q$	$\neg Q \to \neg P$	$\neg(P \wedge \neg Q)$
T	T	F	F	T	T	T
T	F	F	T	F	F	F
F	T	T	F	T	T	T
F	F	T	T	T	T	T

The last three columns are identical. So proving any one of them proves the other two, and you may pick whichever is easiest to attack:

(1) $P \to Q$ — assume P, produce Q;

(2) $\neg Q \to \neg P$ — assume $\neg Q$, produce $\neg P$;

(3) $\neg(P \land \neg Q)$ — assume both P and $\neg Q$, produce a contradiction.

Nothing distinguishes them logically. They differ only in what lands in your hands when you start, and that is a practical matter, not a matter of truth.

The three skeletons

Written as programs, before any of the bodies are known. The indentation is not decoration: a line is available exactly to the lines indented under it, and a block is closed by the line that returns to the outer margin. The bracket is the hole you still owe.

Direct

Let $n \in \mathbb{Z}$ be arbitrary.

> Suppose n is even.
>
> > [Proof that $n+3$ is odd goes here.]
>
> Thus, if n is even then $n+3$ is odd.

Thus, for all $n \in \mathbb{Z}$, if n is even then $n+3$ is odd.

Contrapositive

Let $n \in \mathbb{Z}$ be arbitrary.

> Suppose $n+3$ is not odd.
>
> > [Proof that n is not even goes here.]
>
> Thus, if $n+3$ is not odd then n is not even.

Thus $\neg Q \to \neg P$ for all n, which is $P \to Q$.

Contradiction

Let $n \in \mathbb{Z}$ be arbitrary.

> Suppose n is even and $n+3$ is not odd.
>
> > [Proof of a contradiction goes here.]

Thus n even and $n + 3$ not odd is impossible.

Thus $\neg(P \wedge \neg Q)$ for all n, which is $P \to Q$.

Write these outer lines first. They are forced by the shape of the statement and the method alone — before you have any idea why the claim is true.

The three proofs

Proof 1 (direct)

Assume P; produce Q.

Need to Show:

Assuming n is even, produce an integer j with $n+3 = 2j+1$. That is the whole obligation: "$n+3$ is odd" is by definition the existence of such a j, so the proof is finished exactly when one has been exhibited.

Proof 1. Let $n \in \mathbb{Z}$ be arbitrary.

Suppose n is even.

$n = 2k$ for some $k \in \mathbb{Z}$. [definition of even]

$n + 3 = 2k + 3$. [add 3 to both sides]

$2k + 3 = 2k + 2 + 1 = 2(k + 1) + 1$. [algebra]

$k + 1 \in \mathbb{Z}$. [closure under addition]

So $n + 3 = 2j + 1$ with $j = k + 1 \in \mathbb{Z}$: $n + 3$ is odd. [definition of odd]

Thus, if n is even then $n + 3$ is odd.

Thus, for all $n \in \mathbb{Z}$, if n is even then $n + 3$ is odd. □

Remark. The witness is $j = k + 1$, and producing it is the entire proof. Notice what was *not* used: neither fact. This argument would survive in a world where integers could be both even and odd, or neither. It is the cheapest of the three, and on a statement this shape it is the one to reach for.

Proof 2 (contrapositive)

Assume $\neg Q$; produce $\neg P$.

Need to Show:

Prove $\neg Q \to \neg P$: if $n+3$ is not odd, then n is not even. Both ends need unfolding before there is anything to do:

(a) the hypothesis "$n+3$ is not odd" is unusable as it stands — it offers no k — and Fact 1 converts it into "$n + 3$ is even", which does;

(b) the conclusion "n is not even" is a negative, and Fact 2 lets it be reached by proving the positive statement "n is odd" instead.

So the work is: assuming $n + 3 = 2m$, produce an integer j with $n = 2j + 1$.

Proof 2. Let $n \in \mathbb{Z}$ be arbitrary.

Suppose $n + 3$ is not odd.

$n + 3$ is even.	[Fact 1: not odd, so even]
$n + 3 = 2m$ for some $m \in \mathbb{Z}$.	[definition of even]
$n = 2m - 3$.	[subtract 3 from both sides]
$2m - 3 = 2m - 4 + 1 = 2(m - 2) + 1$.	[algebra]
$m - 2 \in \mathbb{Z}$.	[closure under subtraction]
So $n = 2j + 1$ with $j = m - 2 \in \mathbb{Z}$: n is odd.	[definition of odd]
n is not even.	[Fact 2: odd, so not even]

Thus, if $n + 3$ is not odd then n is not even.

Thus $\neg Q \to \neg P$ holds for every $n \in \mathbb{Z}$; equivalently, for every $n \in \mathbb{Z}$, if n is even then $n + 3$ is odd. [contraposition]

□

Remark. The last line is not a flourish — it is the step that converts what was proved into what was claimed, and omitting it leaves the proof formally incomplete. It is also the step students most often skip, because by that point the arithmetic feels finished. It isn't: the arithmetic established $\neg Q \to \neg P$, and only the truth table in *Why there are three methods at all* turns that into $P \to Q$.

Proof 3 (contradiction)

Assume P and $\neg Q$; produce a contradiction.

Need to Show:

Assuming both that n is even and that $n + 3$ is not odd, derive a statement of the form R and $\neg R$. Nothing specifies which R — any contradiction will do, and finding one is the creative part. Here the target will be a violation of Fact 2: an integer exhibited as both even and odd.

Proof 3. Let $n \in \mathbb{Z}$ be arbitrary.

Suppose n is even and $n + 3$ is not odd.

$n = 2k$ for some $k \in \mathbb{Z}$. [definition of even]

$n + 3$ is even. [Fact 1: not odd, so even]

$n + 3 = 2m$ for some $m \in \mathbb{Z}$. [definition of even]

$2k + 3 = 2m$. [substituting $n = 2k$]

$3 = 2m - 2k = 2(m - k)$. [algebra]

$m - k \in \mathbb{Z}$, so 3 is even. [closure; definition of even]

$3 = 2 \cdot 1 + 1$ and $1 \in \mathbb{Z}$, so 3 is odd. [definition of odd]

So 3 is both even and odd, contradicting Fact 2. $\rightarrow\leftarrow$

Thus n even and $n + 3$ not odd is impossible.

Thus $\neg(P \wedge \neg Q)$ holds for every $n \in \mathbb{Z}$; equivalently, for every $n \in \mathbb{Z}$, if n is even then $n + 3$ is odd. □

Remark. Two hypotheses enter and both get used — $n = 2k$ and $n + 3 = 2m$ — and subtracting one from the other is what makes n disappear, leaving a false statement about the constant 3. That is characteristic: a contradiction proof often ends far from where it started, on a claim about something the original statement never mentioned.

Comparison

	Assume	Produce	Background used
Direct	P	Q	—
Contrapositive	$\neg Q$	$\neg P$	Facts 1 and 2
Contradiction	$P \wedge \neg Q$	$\rightarrow\leftarrow$	Facts 1 and 2

Three observations worth making out loud.

(1) *The proofs are not equally cheap.* The direct proof spends nothing but the definitions. The other two spend both facts, because each begins with a negative hypothesis and a negative hypothesis gives you nothing to compute with until it has been turned positive. On this claim, then, the direct proof is simply the right one; the others are run here for practice.

(2) *Contradiction hands you the most and demands the least.* You may assume P and $\neg Q$ both, and any absurdity discharges the obligation. That generosity is why it is the natural default when the conclusion is itself a negative or an impossibility — as in the lexicographic argument later in the course, where the conclusion is that no utility representation exists and there is nothing to construct. The same generosity

is why it is overused: an argument that could have been direct is often written by contradiction out of habit, with the second hypothesis never touched. If you finish a proof by contradiction without having used $\neg Q$, you have written a direct proof inside a wrapper — delete the wrapper.

(3) *The closing line is part of the proof.* Methods 2 and 3 each end by converting what was established into what was claimed. That conversion is licensed by the truth table, not by the algebra, and a write-up that stops at the algebra has proved the wrong statement.

Remark. A useful exercise, and the natural next one: run the same three methods on *if* n^2 *is odd, then* n *is odd.* There the asymmetry is much sharper — the contrapositive proof is three lines, and the direct proof is genuinely awkward. Comparing the two examples is what makes the point that choosing a method is a real decision rather than a formality.